

Oracle Cloud Computing

Objectives

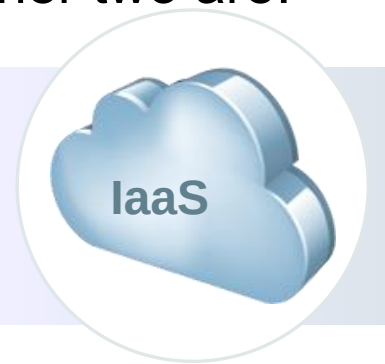
After completing this lesson, you should be able to:

- Define Infrastructure as a Service (IaaS)
- Describe Oracle Private Cloud Appliance
- Describe Oracle OpenStack
- List some Oracle Cloud Infrastructure services
- Describe some key concepts and terms used in Oracle Cloud Infrastructure
- List some Oracle-provided images and available shapes in Oracle Cloud Infrastructure
- Describe the task flow to launch an Oracle Cloud Infrastructure instance
- Describe creation of an Oracle Cloud Infrastructure Virtual Cloud Network and subnet
- Describe oci-utils
- Describe attachment of an Oracle Cloud Infrastructure block volume to an instance



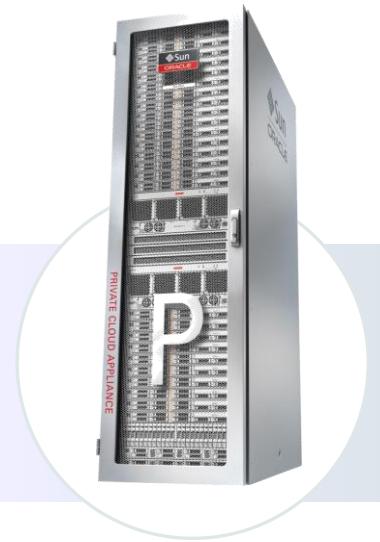
What Is Infrastructure as a Service (IaaS)?

- It is a form of cloud computing that provides infrastructure services over the internet.
- It is required by all applications, databases, and middleware deployments.
- Users can access:
 - Computer processors
 - Storage
 - Networks
 - Other infrastructure resources
- IaaS is one of the three main “service models” in cloud computing. The other two are:
 - SaaS: Software as a Service
 - PaaS: Platform as a Service



What Is Oracle Private Cloud Appliance?

- It is preconfigured for stability, high availability, and automation.
- Orchestration software automatically handles new server hardware.
- It supports the provisioning of Infrastructure and Platform as a Service (IaaS and PaaS) on demand.
- Private Cloud Appliance Virtualization
 - Focus on services, not managing hardware
 - Created to handle demanding workloads
 - Unified support for the software solution stack
- Private Cloud Appliance simplifies application deployment and management.



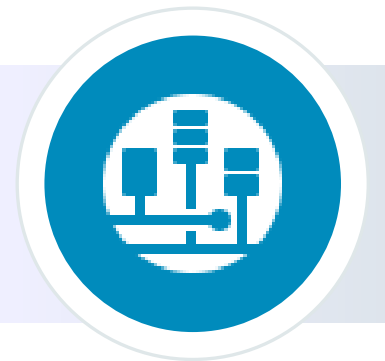
What Is Oracle OpenStack?

- Oracle OpenStack is a cloud management software for managing large pools of compute, storage, and networking resources.
- Based on an OpenStack community release, Oracle OpenStack is an enterprise-grade solution for managing an entire IT environment.
- It rapidly deploys Oracle and third-party applications across shared compute, network, and storage resources.
- It accelerates application deployment with self-service VM creation.

What Are Oracle Cloud Infrastructure Services?

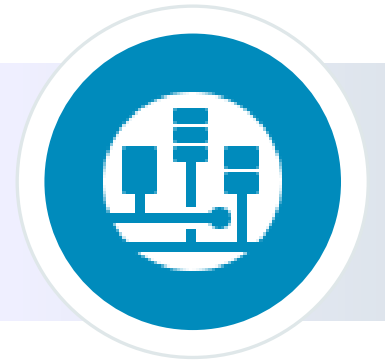
A set of complementary cloud services that enable you to build and run a wide range of applications and services in a highly-available hosted environment. Examples include:

- **Compute Service:** Provision and manage bare metal compute instances or virtual machine instances.
- **Networking Service:** Create and manage the network components for your cloud resources.
- **Identity and Access Management (IAM) Service:** Control access to cloud resources.
- **Load Balancing Service:** Create a public or private load balancer within your virtual cloud network (VCN).



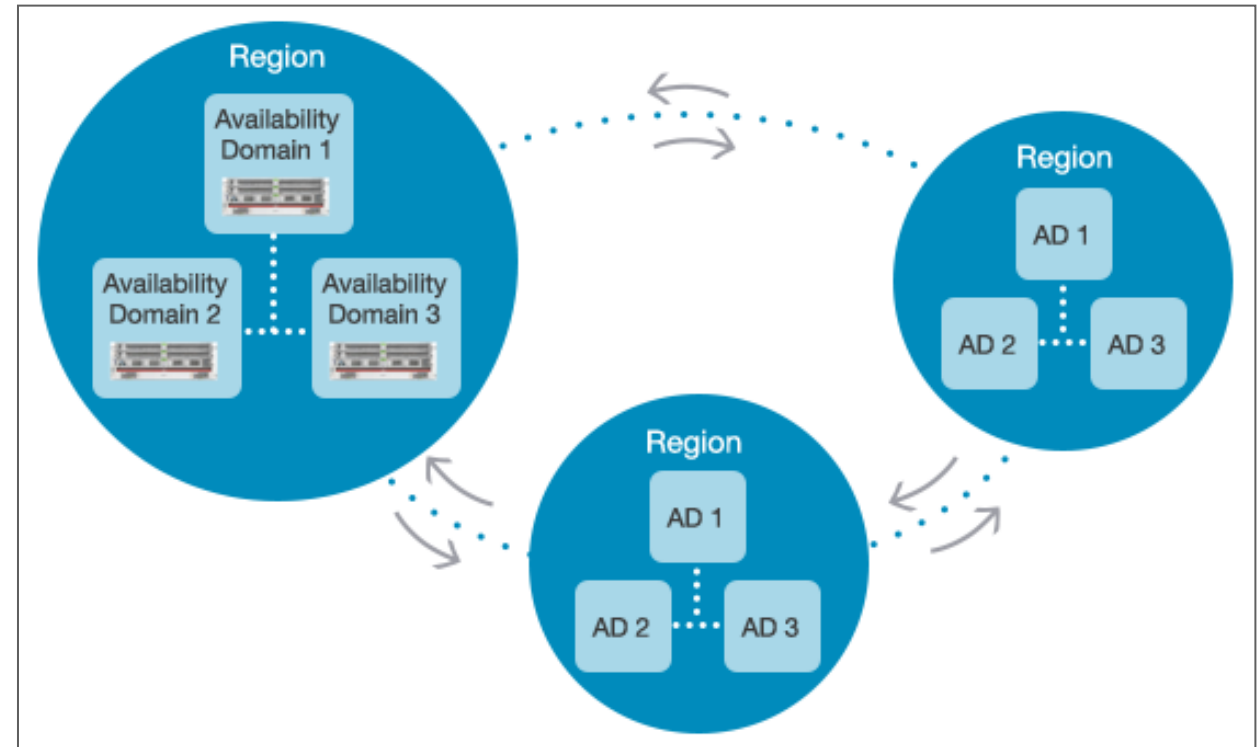
What Are Oracle Cloud Infrastructure Services?

- **Block Volume Service:** Dynamically provision and manage block storage volumes.
- **Object Storage Service:** Manage data as objects stored in logical containers called buckets. There are two storage class tiers:
 - Standard Object Storage (for data needing fast, immediate frequent access)
 - Archive Storage (for data that is seldom or rarely accessed)
- **File Storage Service:** Provide a durable, scalable, distributed, enterprise-grade network file system.
- **Database Service:** Provision a user-managed DB system or autonomous database.
- **Audit Service:** Track activity in your environment.



Key Concepts and Terms

- Regions and Availability Domains
 - A region is a localized geographic area composed of several Availability Domains.
 - An Availability Domain is one or more data centers located within a region.
 - Availability Domains are isolated from each other, fault tolerant, and very unlikely to fail simultaneously.
 - A logical grouping
- Fault Domain
 - A logical grouping of hardware and infrastructure within an availability domain



Key Concepts and Terms

- Tenancy
 - A tenancy is a secure and isolated partition within Oracle Cloud Infrastructure where you can create, organize, and administer your cloud resources.
 - Oracle creates a tenancy for your company when you sign up for Oracle Cloud Infrastructure.
 - Your tenancy is the root compartment that holds all your cloud resources.
- Compartments
 - Compartments allow you to organize and control access to your Oracle Cloud Infrastructure resources.
 - You can create additional compartments within the tenancy (root compartment).
 - When you create an Oracle Cloud Infrastructure resource, you must specify to which compartment you want the resource to belong.

Key Concepts and Terms

- Console
 - A web-based user interface
 - Provides access to manage Oracle Cloud Infrastructure
 - A navigation menu lets you access service pages to create, manage, and view cloud resources
 - A Help menu provides links to documentation and Oracle Support
- Tags
 - Defined keys and values that are associated with resources

Key Concepts and Terms

- Instance
 - An instance is a compute host running in the cloud.
 - Bare metal compute instances run on bare metal servers without a hypervisor.
 - You maintain sole control of the physical CPU, memory, and Network Interface Card.
 - You do not share the physical machine with any other tenants.
 - Managed Virtual Machine (VM) instances are also available for workloads that don't require dedicated physical servers or the high performance of bare metal instances.
- Image
 - The image is a template of a virtual hard drive that defines the operating system and other software for an instance (for example, Oracle Linux).
 - When you launch an instance, you define its characteristics by choosing its image.
- Shape
 - The shape specifies the number of CPUs and the amount of memory allocated to an instance, and specifies if the instance is a bare metal instance or a VM instance.

Oracle-Provided Images

Image	Name	Description
Oracle Linux 7 Unbreakable Enterprise Kernel Release 4	Oracle-Linux-7.x-<date>-<number>	The Unbreakable Enterprise Kernel (UEK) is Oracle's optimized operating system kernel for demanding Oracle workloads. GPU shapes are supported with this image.
Oracle Linux 6 Unbreakable Enterprise Kernel Release 4	Oracle-Linux-6.x-<date>-<number>	The Unbreakable Enterprise Kernel (UEK) is Oracle's optimized operating system kernel for demanding Oracle workloads.
CentOS 7	CentOS-7-<date>-<number>	CentOS is a free, open-source Linux distribution suitable for use in enterprise cloud environments. For more information, see https://www.centos.org/ ↩.

Oracle-Provided Bare Metal Shapes

- Bare Metal Shapes

Shape	Instance Type	OCPU	Memory (GB)	Local Disk	Network Bandwidth ¹	Max VNICS Total: Linux	Max VNICS Total: Windows
BM.Standard2.52	X7-based standard compute capacity	52	768	Block storage only	2 x 25 Gbps	52 total (26 per physical NIC)	27 total (1 on the first physical NIC, 26 on the second)
BM.DenseO2.52	X7-based dense I/O compute capacity	52	768	51.2 TB NVMe SSD (8 drives)	2 x 25 Gbps	52 total (26 per physical NIC)	27 total (1 on the first physical NIC, 26 on the second)

Oracle-Provided VM Shapes

- VM Shapes

Shape	OCPU	Memory (GB)	Local Disk (TB)	Network Bandwidth ¹	Max VNICs Total: Linux	Max VNICs Total: Windows
VM.Standard2.1	1	15	Block Storage only	1 Gbps	2	2
VM.Standard2.2	2	30	Block Storage only	2 Gbps	2	2
VM.Standard2.4	4	60	Block Storage only	4.1 Gbps	4	4

Key Concepts and Terms

- Virtual Cloud Network (VCN):
 - Is a virtual version of a traditional network on which your instances run
 - Includes subnets, route tables, and gateways
 - Resides within a single region, but subnets can belong to multiple availability domains
 - Can have an optional internet gateway to handle public traffic
 - Can have an optional IPSec VPN connection to securely extend your on-premises network
- Subnet
 - Subdivision of a cloud network
 - Exists in a single availability domain
 - Consists of a contiguous range of IP addresses that do not overlap with other subnets in the cloud network

Task Flow to Launch an Oracle Cloud Infrastructure Instance

1. Create an SSH key pair.
2. Create or choose a compartment for your resources.
3. Create or choose a virtual cloud network (VCN).
4. Create or choose a subnet for the VCN.
5. Create an instance.
6. Connect to your instance.
7. Provision and manage block storage volumes (optional).
 - a) Add a block storage volume.
 - b) Attach the volume to an instance.
 - c) Connect a volume to an instance's guest OS.

Setting Up a Virtual Cloud Network (VCN)

- Navigate to the interface for creating a VCN.
- Select an appropriate compartment.
- You can optionally provide a friendly name.
- A CIDR block range must be specified, which cannot later be changed.
- You can specify a DNS label for the VCN.
- If you create the VCN only, you must specify a CIDR block and subsequently create:
 - At least one subnet
 - An internet gateway
 - A route rule
- You can create the VCN with related resources, which automatically provides a CIDR block, subnets, a gateway, and a route rule for you.

Setting Up VCN Subnets

- Navigate to the interface for creating a VCN subnet.
- You can optionally provide a friendly name.
- An availability domain must be selected.
- A CIDR block range must be specified, which cannot later be changed.
- By default, subnets are considered public, which is required to access the internet.
 - You can change this to private
 - This choice cannot later be changed.
- You can optionally specify a route table, one or more security lists, and DHCP options.
 - These can be changed at any time.
- If compartment selection is enabled, you specify the compartment where you want the subnet to reside, as well as for the route table, security lists, and DHCP options.

Viewing Instance Details and Accessing Your Instance

- Navigate to view instance details.
- You can view instance information:
 - Availability domain, fault domain, compartment
 - Shape, image, VCN
- VNIC information is provided:
 - IP addresses
 - FQDN
 - Subnet
- Any attached block volumes are shown.
- Log in to an instance using a public IP address and private key.

OCI Utilities / `oci-utils`

- Preinstalled set of utilities to make it easier to work with Oracle Linux images
- Service component
 - `ocid`
 - Normally runs as a daemon started via `systemd`.
- Command line tools
 - `oci-growfs`: Expands the root file system of the instance to its configured size
 - `oci-iscsi-config`: Displays and configures iSCSI devices attached to a compute instance
 - `oci-metadata`: Displays metadata for the compute instance
 - `oci-network-config`: Lists or configures Virtual Network Interface Cards (VNICs) attached to the compute instance
 - `oci-network-inspector`: Displays a detailed report for a given compartment or network
 - `oci-public-ip`: Displays the public IP address of the current system in either human-readable or JSON format

Creating a Block Volume

- Navigate to the interface for creating a block volume.
- Select a compartment.
- Provide a friendly name.
- An availability domain must be selected and be the same as for the instance.
- Select a size in 1 GiB increments.
- A backup policy can be selected.
- Tags can optionally be applied.
- You can specify that data be encrypted.

Attaching a Block Volume to an Instance

- Navigate to the interface for attaching a block volume to an instance.
- Select the iSCSI (the default) or Paravirtualized attachment type.
- Select a compartment.
- Select the block volume desired.
- CHAP credentials can optionally be enabled.
- Select READ/WRITE or READ-ONLY for the access type.

Connecting a Block Volume to an Instance's Guest OS

- For iSCSI attachments, iSCSI commands must be issued to connect the volume to the OS.
- The Oracle Cloud Infrastructure console provides the necessary iSCSI commands.
- Issue the given commands on your instance.
- Alternatively, run the `oci-iscsi-config` utility from the OCI utilities.
- If the `ocid` daemon is started, devices are connected automatically.
- You can then partition the new block device, create a file system on the device, create a mount point, and mount the new file system.
- Include the `_netdev` and `nofail` options on every nonroot block volume in the `/etc/fstab` file.

Oracle Cloud Computing Resources

Refer to the following resources for further information about Oracle Cloud Computing:

- <https://www.oracle.com/cloud>
- <https://cloud.oracle.com/home>
- <https://cloud.oracle.com/cloud-infrastructure>
- <https://docs.cloud.oracle.com>
- <https://education.oracle.com/oracle-cloud-learning-subscriptions>

Quiz



Which of the following statements are true? (Select all that apply.)

- a. Oracle Cloud Infrastructure is physically hosted in regions and Availability Domains.
- b. Oracle Cloud Infrastructure is physically hosted in tenancies and compartments.
- c. When you sign up for Oracle Cloud Infrastructure, Oracle creates a tenancy for your company.
- d. Compartments allow you to organize and control access to your cloud resources.



Quiz



Which of the following statements about Oracle Cloud Infrastructure instances are true? (Select all that apply.)

- a. You need to set up at least one Virtual Cloud Network (VCN) before you can launch instances.
- b. Bare metal compute instances run on bare metal servers without a hypervisor.
- c. Managed Virtual Machine (VM) instances are also available for workloads that don't require dedicated physical servers or the high performance of bare metal instances.
- d. When launching an instance, you also need to select an image, a shape, a private IP address, and a public IP address.



Summary

In this lesson, you should have learned how to:

- Define Infrastructure as a Service (IaaS)
- Describe Oracle Private Cloud Appliance
- Describe Oracle OpenStack
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Practice 3: Overview

This practice covers Quiz – Oracle Cloud Computing.